Tesla Powerwall Tuning Notes

2 units installed December 2025 - paired with 8.4 kW solar - still being tuned

Goal

Maximize self-consumption of solar and shave the evening peak without leaving the house exposed during a winter grid outage. Balance backup reserve against time-of-use savings.

Current Settings

Parameter	Setting	Note
Mode	Time-Based Control	Switched from Self-Powered in January
Backup reserve	30 percent	Raised from 20 after a Jan cold snap
Peak window	4 PM to 9 PM	Matches utility TOU
Storm Watch	Enabled	Auto-charges ahead of severe weather

Observations

At 30 percent reserve the system still covers the evening load most days from solar surplus. On the two coldest January nights the heat pump pulled the pack below reserve by 6 AM and the grid topped up at off-peak rates, which is acceptable. Summer plan, when AC runs April to October, is to drop reserve to 20 percent and lean harder on peak shaving.

Daily State of Charge Log

Readings taken at the 9 PM close of the peak window and again at the 6 AM low point. Values are pulled from the local gateway API into Home Assistant.

Date	9 PM SoC	6 AM SoC	Grid topup	Note
2026-02-03	64 percent	41 percent	No	Mild night
2026-02-08	58 percent	33 percent	No	Cloudy afternoon
2026-02-11	71 percent	46 percent	No	Strong solar day
2026-02-14	39 percent	30 percent	Brief	Cold snap, heat pump
2026-02-19	66 percent	44 percent	No	Typical
2026-02-24	69 percent	45 percent	No	Sweet spot ride day

Round-Trip Efficiency

Across February the pack absorbed 210 kWh and discharged 198 kWh, a round-trip efficiency near 94 percent, consistent with the paired inverter losses and the cool ambient temperature in the garage.

Seasonal Reserve Plan

Season	Months	Reserve	Priority
Winter	November to March	30 percent	Outage resilience
Shoulder	April, October	25 percent	Balanced
Summer	May to October	20 percent	Peak shaving

Integration Roadmap

- Publish Powerwall state of charge, mode, and reserve into Home Assistant as first-class sensors.
- Automate the seasonal reserve schedule so the change is hands-off.
- Build a custom energy dashboard that overlays solar, load, and pack state.
- Add a notification when reserve is breached before 6 AM two nights running.

Event Log

Date	Event
2025-12-14	Two units commissioned and paired with the inverter
2026-01-06	Switched from Self-Powered to Time-Based Control
2026-01-15	Raised reserve to 30 percent after cold snap
2026-02-01	Storm Watch pre-charge ahead of a front
2026-02-20	Storm Watch pre-charge, second event of the season

Next

Integrate Powerwall state of charge into Home Assistant for a custom dashboard and automate a seasonal reserve schedule. Revisit after the first full summer cycle. Target review date is October 2026.